STA301-Statistics and Probability

Final term Current Papers (Fall 2022)

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1. **The parameter of the chi-square distribution is**

    A. V Correct
    B. V-1
    C. V-2
    D. V^{-P}

2. **The Hyper geometric random variable is a**

    A. Continuous Variable
    B. Discrete Variable Correct
    C. Undefined
    D. Independent Variable

3. **The Degree of freedom for a t-test with sample size 10 is.**

    A. 5
    B. 8
    C. 9
    D. 10 Correct

4. **The Degree of freedom for a t-test with sample size 6 is.**

   A. 1
   B. 3
   C. 5 Correct
   D. 7

5. **When C is constant then E(C) is**

   A. 0
   B. 1
   C. C Correct (Page no 180)
   D. -C

6. **When the random variable X and Y are independ then their co-variance is.**

   A. One
   B. Negative
   C. Zero Correct
   D. Positive

7. **In normal distribution the quartile deviation Q.D = ____________.**

   A. 0.5 σ
   B. 0.75 σ
   C. 0.7979 σ
   D. 0.6745 σ Correct

8. **To find the confidence interval for the ratio of two variances, we use**

   A. F-Distribution Correct
   B. Z-Distribution
   C. Chi-square Distribution
   D. t-Distribution

9. F-Distribution is

- **A. Positively Skewed   Correct (Page no 312)**
- B. Negatively Skewed
- C. Normal
- D. Symmetrical

10. In case of a of a 3 X 3 contingency table what is the value of degrees of freedom?

- A. 2
- B. 4
- C. 6
- **D. 9 Correct**

11. What is the graphical shape of the chi-square distribution

- **A. Positively skewed   Correct (Page no 307)**
- B. Negatively skewed
- C. Uniformly distributed
- D. Normally distributed

12. For degree of freedom $V \leq 2$ the variance of t-distribution

- A. is greater than zero
- B. is less than one
- C. is equal to one
- **D. does not exist   Correct (Page no 294)**

13. **When f(x) is continuous probability function then P(x=2) is**

    A. 1
    B. 0.5
    C. **0 Correct**
    D. 0.25

14. **Uniform distribution is defined by**

    A. Largest value
    B. **Largest and smaller value Correct**
    C. smallest value
    D. central value

15. **The end point of a confidence interval called**

    A. Confidence Coefficient
    B. Confidence interval
    C. **Confidence limits Correct**
    D. Parameters

16. **A 95% confidence interval for population proportion P is 32.4 % to 47.6% the value of sample proportion is**

    A. **40% Correct**
    B. 32.4%
    C. 47.6%
    D. 80%

17. **In a bivariate probability distribution of X and Y if E(X)=3.2 and E(Y)=3.0 , then E(X+Y)**

 A. **6.2 Correct**
 B. 6.8
 C. 0.2
 D. 0

18. **In testing of hypothesis, a + b is always equal**

 A. **One Correct**
 B. Two
 C. Zero
 D. Difficult to tell

Subjective

Question no 1:

A manufacturer can batteries claims that the life his batteries has a standard deviation equal to 0.9 year. if a random sample of 10 of these batteries have a standard deviation of 1.2 year, do you think that 0 < 0.9 years? use a 0.05 level of significance.

Answer:

To determine whether the sample standard deviation of 1.2 years is significantly different from the population standard deviation of 0.9 years, we can use a one-tailed t-test with a significance level of 0.05.

The null hypothesis is that the population standard deviation is equal to 0.9 years:

H0: σ = 0.9

The alternative hypothesis is that the population standard deviation is greater than 0.9 years:

Ha: σ > 0.9

The test statistic for the one-tailed t-test is given by:

t = (s / sqrt(n-1)) / (σ0 / sqrt(n))

where s is the sample standard deviation, n is the sample size (which is 10 in this case), and σ0 is the hypothesized value of the population standard deviation under the null hypothesis.

Plugging in the given values, we get:

t = (1.2 / sqrt(10-1)) / (0.9 / sqrt(10)) ≈ 1.32

The degrees of freedom for the t-test is (n-1) = 9.

Using a t-table with 9 degrees of freedom and a significance level of 0.05 for a one-tailed test, we find that the critical value is 1.833.

Since our calculated t-value of 1.32 is less than the critical value of 1.833, we fail to reject the null hypothesis.

Therefore, at the 0.05 level of significance, we do not have enough evidence to conclude that the population standard deviation is greater than 0.9 years. In other words, we cannot say that 0.9 years is too small for the standard deviation of the battery life.

Question no 2:

Can we reject a claim that the average weight of a table of a certain drug is claimed to be 50mg.if a random sample of 100 tablets showed a mean weight of 50.15 mg with a standard deviation of 0.4 mg. Assume that all the tablets are normally distributed

Where (Z_x= 258)

Answer:

To determine whether we can reject the claim that the average weight of a tablet of a certain drug is 50 mg, we can perform a one-sample t-test with a significance level of 0.05.

The null hypothesis is that the true population mean is equal to 50 mg:

$H0: \mu = 50$

The alternative hypothesis is that the true population mean is greater than 50 mg:

$Ha: \mu > 50$

The test statistic for the one-sample t-test is given by:

$t = (\bar{x} - \mu 0) / (s / \mathrm{sqrt}(n))$

where $\bar{x}$ is the sample mean, $\mu 0$ is the hypothesized value of the population mean under the null hypothesis, s is the sample standard deviation, and n is the sample size.

Plugging in the given values, we get:

$t = (50.15 - 50) / (0.4 / \mathrm{sqrt}(100)) \approx 3.875$

The degrees of freedom for the t-test is (n-1) = 99.

Using a t-table with 99 degrees of freedom and a significance level of 0.05 for a one-tailed test, we find that the critical value is 1.660.

Since our calculated t-value of 3.875 is greater than the critical value of 1.660, we reject the null hypothesis.

Therefore, at the 0.05 level of significance, we have enough evidence to conclude that the true population mean weight of the tablet of the certain drug is greater than 50 mg. Note that the value of Z_x=258 mentioned in the question does not appear to be relevant to this problem, so we can ignore it.

Question no 3:

A population consists of N=4 values 4,6,8,9. A sample size of n = 3 is selected from the population without replacement. calculate mean and variance of proportion for odd numbers.

Answer:

The proportion of odd numbers in the population is p = 2/4 = 0.5.

To calculate the mean and variance of the proportion of odd numbers in a sample of size n=3 selected without replacement from the population, we can use the hypergeometric distribution.

The probability of selecting k odd numbers in a sample of size n is given by:

P(k odd numbers) = (number of ways to select k odd numbers) * (number of ways to select n-k even numbers) / (number of ways to select any n numbers)

The number of ways to select k odd numbers from the 2 odd numbers in the population is given by:

(number of ways to select k odd numbers) = (number of ways to select k items from 2) = 2Ck

Similarly, the number of ways to select n-k even numbers from the 2 even numbers in the population is given by:

(number of ways to select n-k even numbers) = (number of ways to select n-k items from 2) = $2C(n-k)$

The number of ways to select any n numbers from the population is given by:

(number of ways to select any n numbers) = (number of ways to select n items from 4) = $4Cn$

Therefore, the probability of selecting k odd numbers in a sample of size n is:

$$P(k \text{ odd numbers}) = (2Ck * 2C(n-k)) / 4Cn$$

We can use this probability distribution to calculate the mean and variance of the proportion of odd numbers in a sample of size n=3 selected without replacement from the population.

The mean of the proportion of odd numbers is given by:

$$\mu = E(p) = \Sigma(k=0 \text{ to } n)\ P(k \text{ odd numbers}) * k/n$$

$$\mu = [(2C0 * 2C3) / 4C3] * 0/3 + [(2C1 * 2C2) / 4C3] * 1/3 + [(2C2 * 2C1) / 4C3] * 2/3 + [(2C3 * 2C0) / 4C3] * 3/3$$

$\mu \approx 0.5$

Therefore, the mean of the proportion of odd numbers in a sample of size n=3 selected without replacement from the population is approximately equal to the proportion of odd numbers in the population.

The variance of the proportion of odd numbers is given by:

$\sigma^2 = Var(p) = \Sigma(k=0 \text{ to } n)\ P(k \text{ odd numbers}) * (k/n - \mu)^2$

$\sigma^2 = [(2C0 * 2C3) / 4C3] * (0/3 - 0.5)^2 + [(2C1 * 2C2) / 4C3] * (1/3 - 0.5)^2 + [(2C2 * 2C1) / 4C3] * (2/3 - 0.5)^2 + [(2C3 * 2C0) / 4C3] * (3/3 - 0.5)^2$

$\sigma^2 \approx 0.056$

Therefore, the variance of the proportion of odd numbers in a sample of size n=3 selected without replacement from the population is approximately equal to 0.056.

Question no 4:

Assume that X has a poison distribution with variance 2. Calculate P(X=2).

Answer:

If X follows a Poisson distribution with a variance of 2, then the parameter λ of the Poisson distribution is given by the variance, which is $\lambda = 2$.

The probability mass function of a Poisson distribution is given by:

$P(X=k) = (e^{(-\lambda)} * \lambda^{k}) / k!$

where k is the number of occurrences, λ is the parameter of the distribution, and e is the mathematical constant $e \approx 2.71828$.

To calculate $P(X=2)$, we substitute $k=2$ and $\lambda=2$ into the formula above:

$P(X=2) = (e^{(-2)} * 2^{2}) / 2! \approx 0.2707$

Therefore, if X follows a Poisson distribution with a variance of 2, then the probability of X taking on a value of 2 is approximately 0.2707.